Practical Data Science Project

Caries Detection by Machine Learning

Dental Carries Surface Classification

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Introduction

This report has been prepared as part of the course **ITDS346 Practical Data Science.** Its primary objective is to study and gain a comprehensive understanding of the process of applying Machine Learning techniques to medical image analysis, which represents one of the key approaches in advancing healthcare innovation in today's digital era. The contents of this report cover the entire workflow, beginning with dental X-ray image preparation, data cleaning, and preprocessing, followed by the development of models for detecting tooth locations and potential caries regions. It also includes model evaluation and performance optimization to ensure the system's suitability for practical applications.

This project focuses on the development of a **Dental Caries Detection system using Machine Learning**, employing Computer Vision and Deep Learning techniques to analyze panoramic dental X-ray images to accurately identify regions at risk of tooth decay. The development process encompasses data management, model construction and training, pixel-level structural analysis of teeth, and visualization of results in a format that can effectively support clinical decision-making in dentistry.

The authors sincerely hope that this report will be beneficial to readers in enhancing their understanding of the application of Data Science and Machine Learning in the healthcare domain. Furthermore, it is intended to serve as a reference or foundation for future research and related projects. Should there be any errors or areas for improvement, the authors would greatly appreciate constructive feedback and sincerely apologize for any shortcomings.

The Authors

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Data Collection & Sources

The dataset used in this study was obtained from the **Faculty of Dentistry, Mahidol University.** It consists of real-world panoramic dental X-ray images along with structured annotation files, which are used as ground truth for model development and evaluation.

Each case in the dataset includes the following components:

- Panoramic Dental X-ray Images: Grayscale images capturing the full dental structure, used as the primary input for the detection and classification pipeline.
- XML Annotation Files (Ground Truth): These files contain structured clinical information, including tooth identifiers based on the FDI notation system and corresponding caries surface labels (Occlusal, Mesial, Distal). These annotations are used as the ground truth for evaluating model performance.
- ROI Mask Images for Caries Regions: Instead of having direct pixel-level annotation in the XML files, caries regions are provided as separate binary mask images. In these images, pixel values greater than zero represent areas affected by caries. These ROI masks are later used to determine the overlap between detected tooth regions and potential caries areas.
- Tooth Polygon / Segmentation Information: Spatial information of each tooth is derived from the segmentation pipeline (YOLO + Detectron2), which provides bounding boxes and pixel-level masks for each tooth instance.

To integrate these data sources, the project performs a mapping process where tooth-level segmentation results are combined with ROI mask images. Specifically, pixel overlap is calculated to determine whether a given tooth contains caries and to estimate the distribution of lesions within that tooth.

All data components are further transformed into structured formats such as JSON to facilitate downstream processing, including surface classification, visualization, and evaluation.

Overall, the dataset provides both imaging data and structured annotations necessary for building a complete pipeline, while requiring additional processing steps to bridge the gap between ROI-based caries representation and tooth-level classification.

Data Preparation

In this project, no traditional data cleaning or preprocessing techniques were explicitly applied. Instead, the data preparation step focuses on **XML parsing and validation of ground truth annotations**.

The XML files were parsed to extract key information, including tooth identifiers and carries surface labels (Occlusal, Mesial, Distal). This parsing process was implemented directly within the evaluation pipeline.

During parsing, each case was validated to ensure that the extracted information was complete and usable. The validation process categorizes data into the following groups:

- **Valid cases**: XML files that contain both tooth ID and surface label
- **Missing tooth information**: Cases where tooth identifiers could not be extracted
- **Missing surface labels**: Cases where surface annotations are incomplete
- **Parsing errors**: XML files that could not be processed due to format issues

Only valid cases were considered in the evaluation stage to ensure the correctness of performance metrics.

In addition to validation, the parsed data was used to generate summary statistics, such as the number of annotated lesions and their distribution across different surface types. These statistics provide an overview of the dataset and help identify potential data issues before model evaluation.

Overall, this step ensures that the ground truth data is reliable and consistent, without introducing additional transformation or restructuring processes.

Dataset Statistics and Exploratory Data Analysis

This section presents both the statistical summary of the dataset and the exploratory analysis used to identify key patterns and potential challenges.

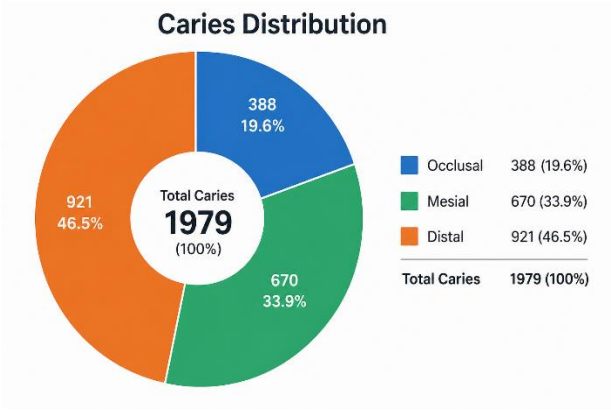
Dataset Statistics

After parsing the XML annotations, basic statistics were computed to summarize the dataset.

The dataset consists of panoramic dental X-ray images with corresponding XML annotations. From the valid parsed data, the following information was obtained:

- Total number of cases
- Number of valid cases used for evaluation
- Total number of annotated caries lesions
- Distribution of surface labels (Occlusal, Mesial, Distal)

Label	Value
Total cases	500
Valid cases	500
Total Caries	1979
Occlusal	388
Mesial	670
Distal	921



Surface Distribution or Dataset Summary

These statistics provide an overview of the dataset structure and ensure that sufficient data is available for analysis.

Exploratory Data Analysis (EDA)

EDA was conducted to identify patterns and issues that may affect model performance.

First, the dataset shows **class imbalance**, where certain surface types (especially Occlusal) appear more frequently than others. This may bias the model toward dominant classes.

Second, **annotation inconsistencies** were observed during XML parsing, such as missing tooth identifiers or incomplete surface labels. These issues reduce the number of usable samples.

Third, **structural variability in dental images** presents a challenge. Teeth appear at different orientations, and some have symmetric shapes, making geometric analysis (e.g., PCA) less reliable.

Overall, these findings highlight that both data quality and structural complexity are key challenges in this task and must be considered in model design and evaluation.

Data Visualization

This section presents both the raw data representation and analytical visualizations used to better understand the dataset and model performance.

Sample Data (Raw Data Representation)

The dataset consists of panoramic dental X-ray images along with structured annotations and ROI masks.

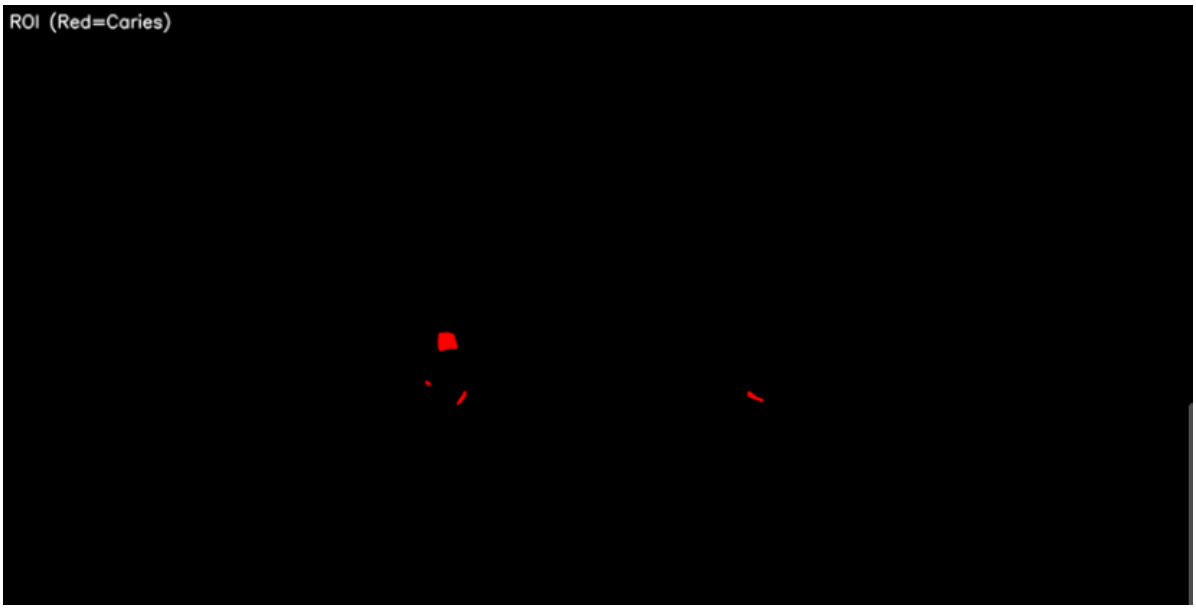
Examples of the raw data include:

- **Panoramic dental X-ray images**, which serve as the primary input

Example Picture of Panoramic dental X-ray image

- **XML annotation files**, containing tooth identifiers and surface labels

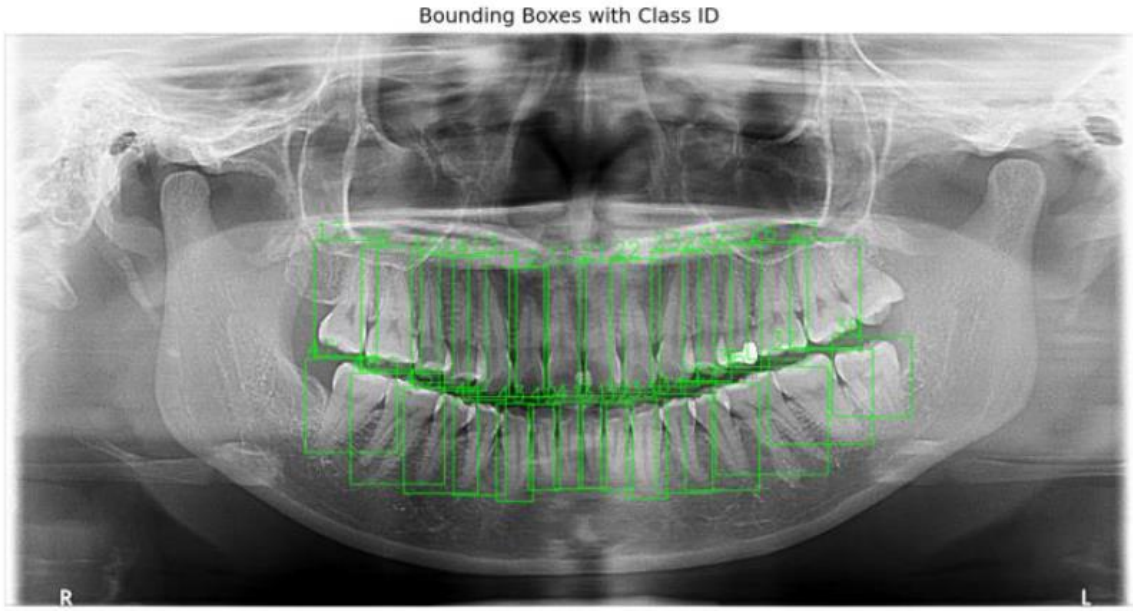
- **ROI mask images**, representing caries regions



Example Picture of ROI mask image

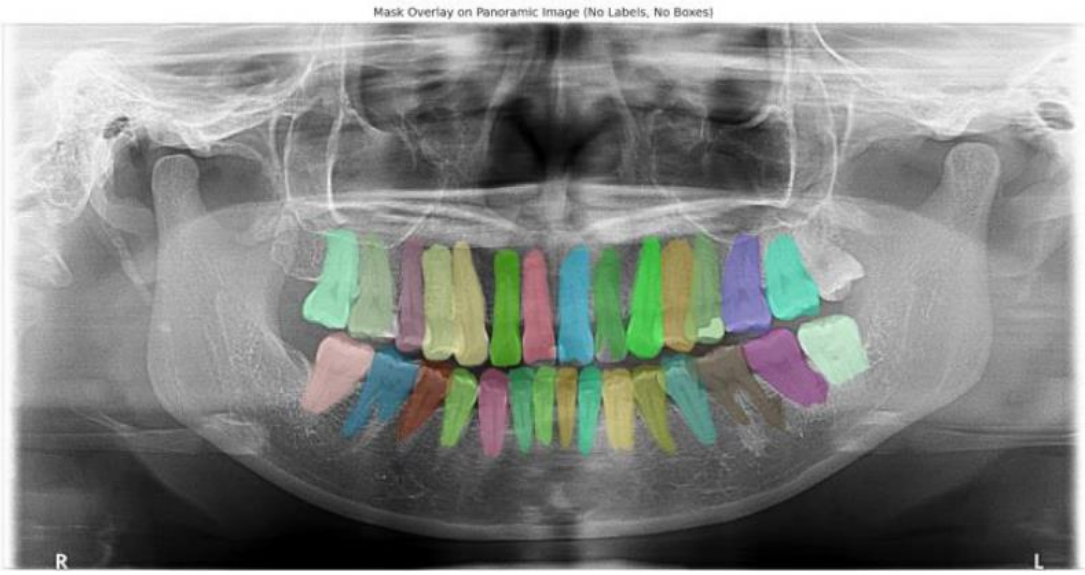
In addition, intermediate outputs from the processing pipeline are visualized, including:

- Tooth detection using bounding boxes



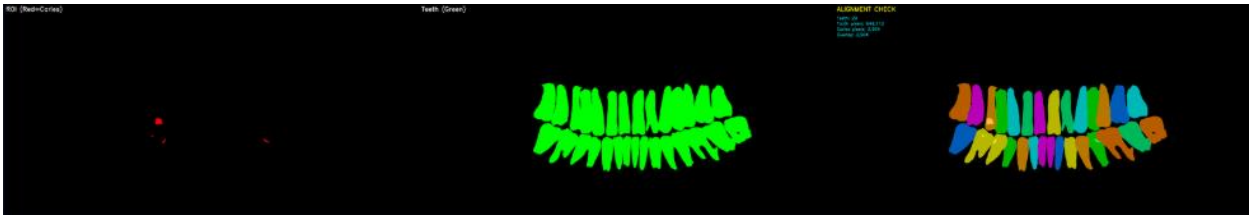
Example Picture of Bounding box of each tooth

- Tooth segmentation using pixel-level masks



Example Picture of Pixel-level tooth mask

- Caries-to-tooth mapping through pixel overlap



Example Picture of caries pixel annotations and tooth masks

These visualizations provide a clear understanding of how raw data is structured and processed in the system.

Analytical Visualization (Insights & Results)

To analyze dataset characteristics and model performance, several visualizations are used.

First, the **distribution of caries surface types** is presented to highlight class imbalance within the dataset.

Second, **confusion matrices** are used to evaluate classification performance by comparing predicted labels with ground truth. This helps identify common misclassification patterns.

Additionally, visualizations from intermediate steps, such as PCA-based orientation and surface zoning, are included to illustrate how the model processes spatial information.

These visualizations support both data understanding and result interpretation.

Feature Engineering & Selection

This section describes the final feature design and processing pipeline used for caries surface classification. The approach focuses on extracting spatial information from lesion distribution and structuring it into interpretable features.

1.Design Overview

The problem is formulated as a **surface classification task**, where the goal is to determine the location of caries (Mesial, Occlusal, Distal) on each tooth.

Instead of relying on learned features from a machine learning model, the system is designed as a **structured pipeline**, where features are explicitly derived from:

- Tooth geometry (segmentation mask)
- Relative position of caries pixels
- Spatial distribution within each tooth

This allows the model to remain interpretable and aligned with clinical reasoning.

2.Tooth-Level Representation

Each tooth is processed independently after detection and segmentation. The following information is extracted:

- Tooth mask (pixel-level)
- Bounding box and spatial coordinates
- Set of pixels representing the tooth region

This representation enables further spatial analysis without requiring additional feature encoding.

3.Coordinate Normalization (Simplified Orientation Handling)

To analyze lesion position consistently across different teeth, a normalization step is applied.

In earlier iterations (Phase 1), multiple PCA-based orientation methods were explored. However, these approaches introduced instability due to:

- Symmetric tooth shapes
- Sensitivity to noise and incomplete masks

In the final design, the system reduces reliance on complex orientation correction and instead focuses on relative spatial distribution within the tooth region. This simplifies the pipeline and improves robustness

.

4.Lesion Distribution-Based Feature Construction

The core feature in the final system is derived from the **distribution of caries pixels within each tooth**.

The process includes:

- Mapping caries pixels onto the tooth region
- Dividing the tooth into predefined zones (Mesial, Occlusal, Distal)
- Measuring the proportion of caries pixels in each zone

To reduce noise:

- Small, isolated components are filtered out
- A minimum threshold (e.g., percentage of pixels) is applied to determine valid region involvement

This approach captures the **spatial extent of the lesion**, rather than relying on a single representative point.

5.Surface Classification Logic

Based on the detected zones, the final surface label is assigned using a rule-based mapping:

- Mesial zone ⟶ M
- Distal zone ⟶ D
- Occlusal zone ⟶ O

This rule-based approach ensures that:

- Predictions are interpretable
- Multi-surface lesions can be handled
- Outputs are consistent with clinical labeling standards

6.Final Design Rationale

The final feature engineering design prioritizes:

- **Robustness**: Less sensitivity to noise and segmentation errors
- **Interpretability**: Clear mapping between spatial features and output labels
- **Simplicity**: Avoiding overly complex geometric transformations

Compared to earlier PCA-heavy approaches, the final pipeline focuses more on **direct spatial analysis of lesion distribution**, which provides more stable and practical results.

Model Selection & Development

The model development in this project is conducted in two main phases, focusing on improving both tooth localization and the accuracy of caries surface classification through geometric analysis.

Phase 1: Tooth Orientation Normalization and Surface Classification

1.PCA-Based Orientation (Selected)

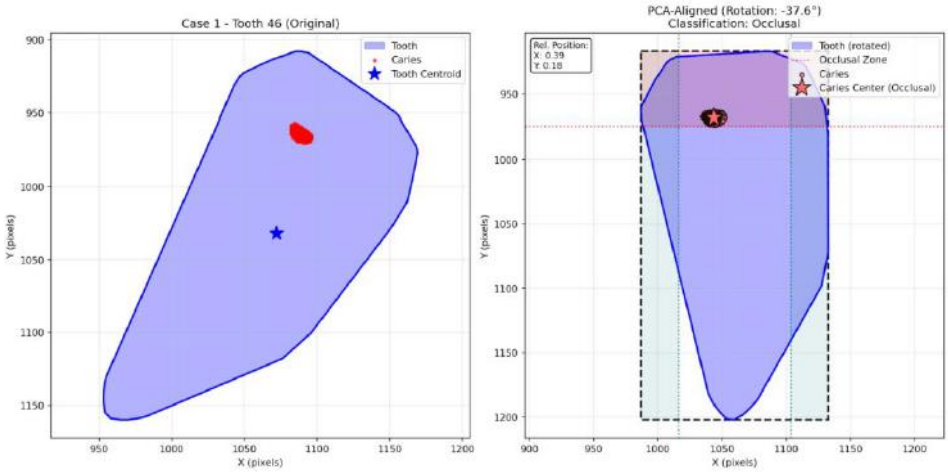
Principal Component Analysis (PCA) was adopted as the core geometric framework for estimating tooth long-axis direction. However, raw PCA was insufficient in many clinical cases, leading to multiple refinement strategies.

PCA Sub-Methods (Total = 7)

(1) Standard PCA (Baseline)

- Compute covariance matrix of mask coordinates.
- Select eigenvector with largest eigenvalue.
- Rotate teeth around centroid.

Limitation: Axis flipping in symmetric or square-shaped molars.



Example Picture of PCA tooth rotation with surface classification

(2) Conditional Ratio

- Measure width-to-height ratio of bounding box.
- If ratio ≈ 1 (square-like), override default eigenvector selection.

Purpose: Reduce 90° mis-rotation in molars.

(3) Maximum Projected Span

- Project mask points onto each eigenvector.
- Compute physical span along each axis.
- Select axis with maximum projected length.

Rationale: Anatomical height should exceed width.

Improvement: More stable than Conditional Ratio.

(4) Split-Centroid Anatomy (Final Selected Orientation)

- Split mask into upper and lower halves.
- Compute centroid of each half.
- Construct anatomical vector from lower (root region) to upper (crown region).
- Select PCA axis best aligned with this vector.

Advantages:

- Encodes biological growth direction.
- Reduces symmetrical ambiguity.

- More resistant to segment noise.
- Achieved highest Macro-F1 in evaluation.

This method was selected as the final phase 1 orientation logic.

(5) Furthest Point Alignment

- Identify boundary point farthest from centroid.
- Use vector toward that point as anatomical reference.

Limitation: Unstable when root apex incomplete. Not selected.

(6) Absolute Vertical Prior

- Select PCA eigenvector closest to global vertical axis of panoramic image.

Assumption: Teeth generally erupt vertically.

Limitation: Sensitive to image tilt. Not robust for full dataset.

(7) Multi-Class Segmentation (Tooth vs Root Separation)

Beyond geometric correction, a structural redesign was proposed:

- Modify segmentation into two classes: Crown, Root
- Compute orientation directly from: Centroid(root) $\longrightarrow$ Centroid(crown)

Advantages:

- Eliminates geometric ambiguity.

- Explicit anatomical axis.
- Theoretically more robust than PCA-only methods.

Limitation: Requires full dataset re-annotation. Not implemented in this phase.

Cloud-Based Surface Classification

After orientation normalization using the Split-Centroid PCA method, surface classification is performed through a cloud-based analysis stage.

The cloud stage consists of two components only:

1. Multi-Zone Point-Cloud Voting
2. Surface Classifier (Rule-Based Label Generator)

1) Multi-Zone Point-Cloud Voting

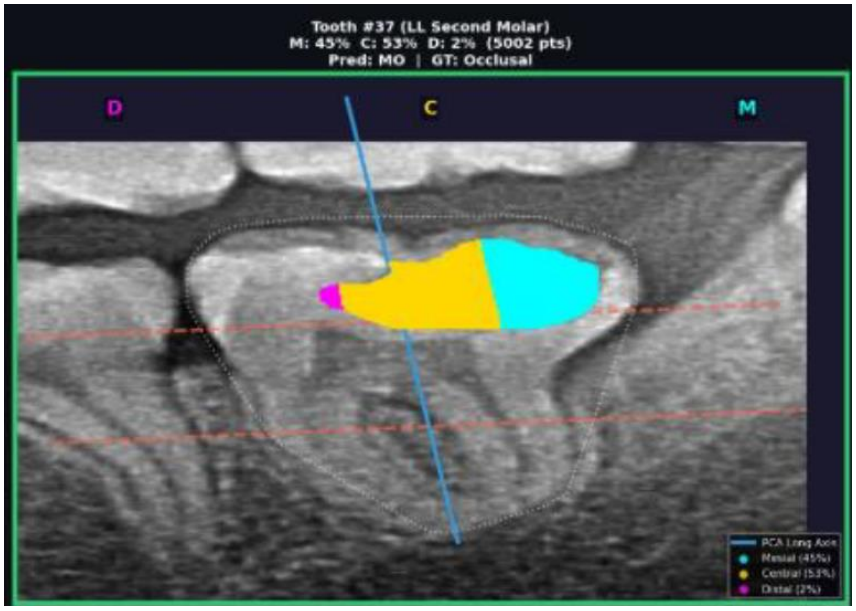
The purpose of this step is to analyze the spatial distribution of lesion pixels within the PCA-normalized coordinate system.

Procedure:

- Rotate all caries pixels into the PCA-aligned coordinate frame.
- Divide the occlusal region into three zones:
    - M (Mesial)
    - C (Central / Occlusal)
    - D (Distal)
- Compute the proportion of lesion pixels within each zone.

- A zone is marked positive if it contains ≥ 5% of total lesion pixels.
- Apply connected-component filtering to remove noise components smaller than 15 pixels.

This approach captures full lesion distribution instead of relying on a single representative point.



Example Picture of Multi-Zone Point-Cloud Voting output

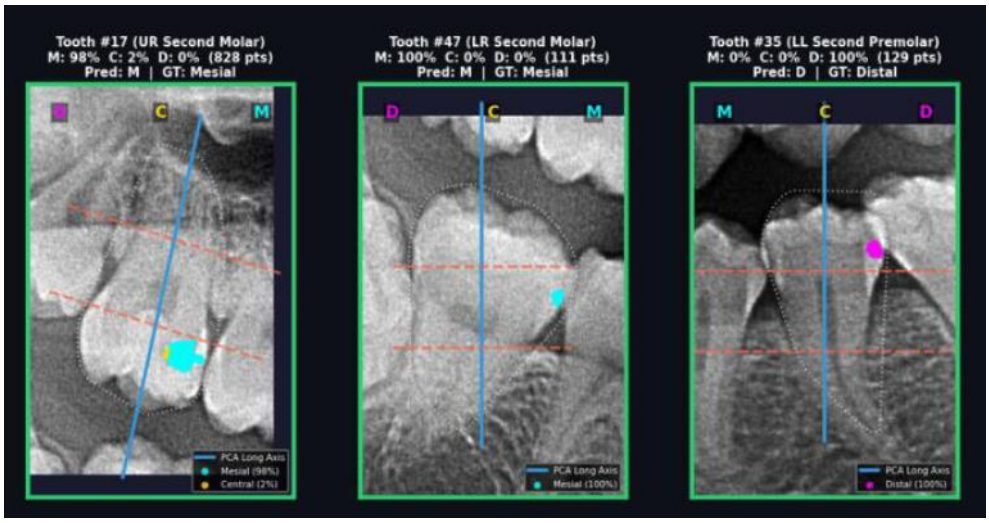
2) Surface Classifier (Rule-Based Label Generator)

After determining zone involvement, the final surface label is assigned using rule-based mapping:

- M only ⟶ M
- D only ⟶ D
- C only ⟶ O
- M + C ⟶ MO
- D + C ⟶ DO

- M + C + D ⟶ MOD

This stage does not introduce an additional machine learning model. It is a deterministic mapping from zone combination to clinically interpretable surface labels.



Example Picture of Surface Classifier output

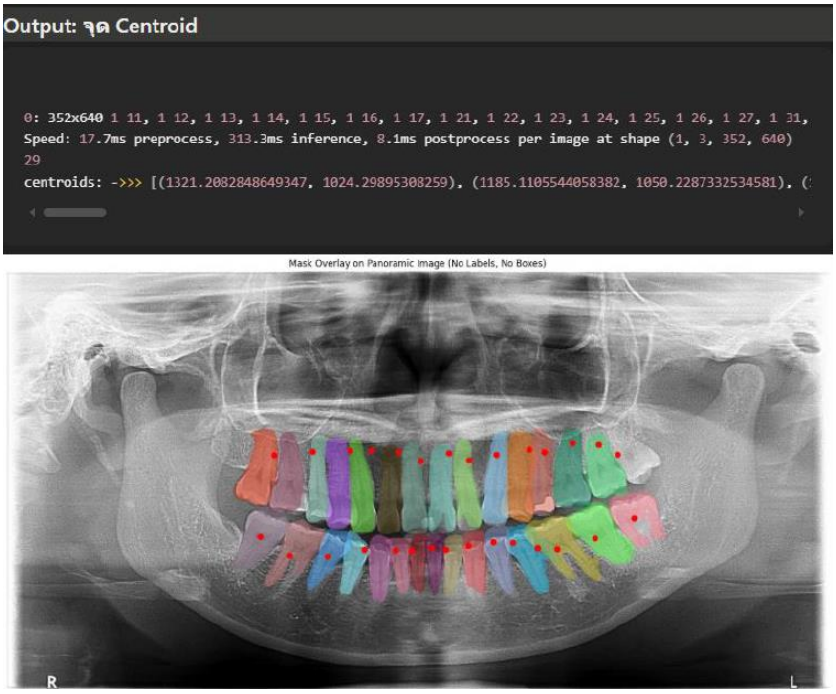
2.Neighbor-Aware Orientation

This method estimated orientation based on alignment of adjacent teeth in the dental arch.

Procedure:

- Compute centroids of neighboring teeth.
- Estimate arch direction.
- Define vertical axis as perpendicular to arch.

Limitation: Incomplete mask availability in JSON stage. Row alignment unstable.



Example Picture of Neighbor-Aware Orientation output

3.Skeleton / Medial Axis-Based Orientation

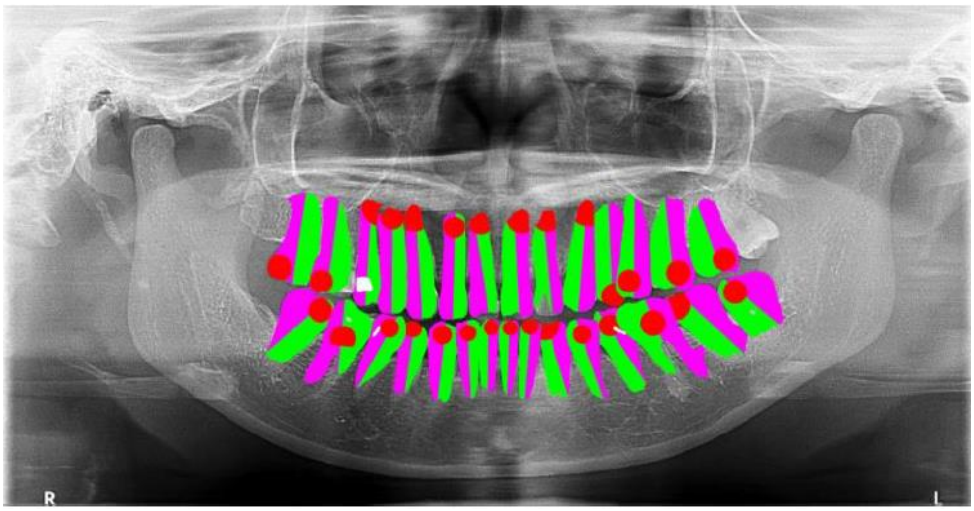
This approach derived anatomical axis from internal structure.

Steps:

1. Convert mask to skeleton.
2. Detect endpoints.
3. Identify thinner endpoints as root.
4. Define axis from root to crown.

Strength: Anatomically intuitive.

Limitation: Highly sensitive to segmentation noise. Unstable for incomplete masks.



Example Picture of Skeleton / Medial Axis-Based Orientation output

Phase 2: Optimization and Model Refinement

The second phase focuses on improving robustness and classification performance by refining both preprocessing and zoning strategies.

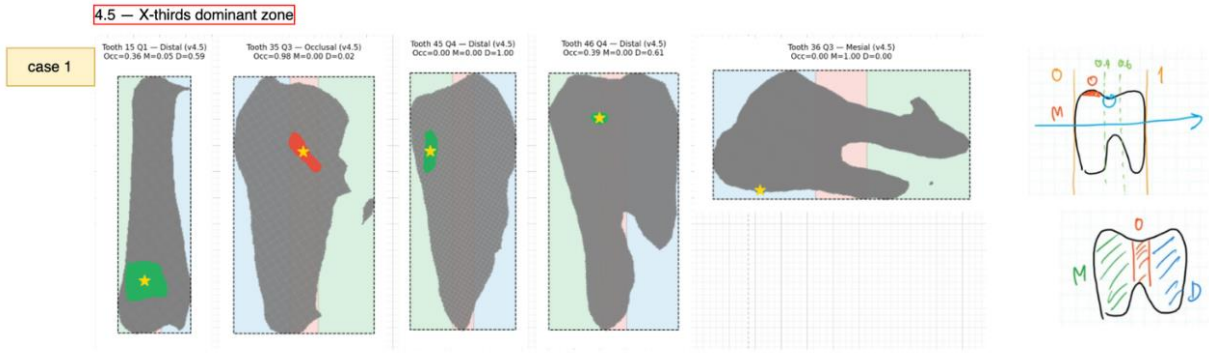
Model Iterations for Surface Classification

Different zoning strategies were explored to determine the most effective method for surface classification:

X-Thirds Dominant Zone (Final Model)

- Divides the tooth into three vertical regions:
    - Mesial: 40%
    - Occlusal: 20%
    - Distal: 40%

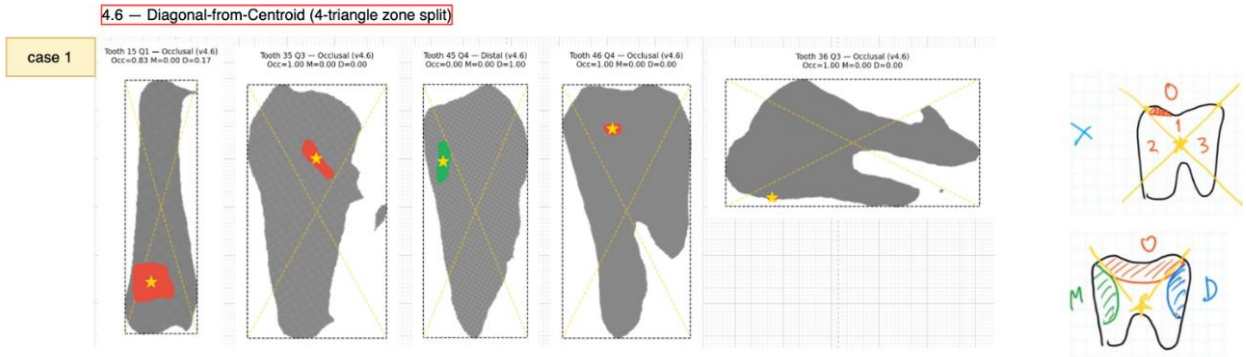
This version achieves the best overall performance and stability.



Example Picture of X-Thirds Dominant Zone output

Diagonal-from-Centroid

- Divides the tooth using diagonal lines into triangular regions
- Improves detection of Occlusal regions
- However, significantly reduces performance for Distal classification



Example Picture of Diagonal-from-Centroid output

Model Evaluation

This section presents the experimental results of all methods developed in this project. The evaluation is organized by method, allowing clear comparison of performance and behavior at each stage.

Phase 1: Geometric PCA-Based Methods

Method 1: Standard PCA (Baseline)

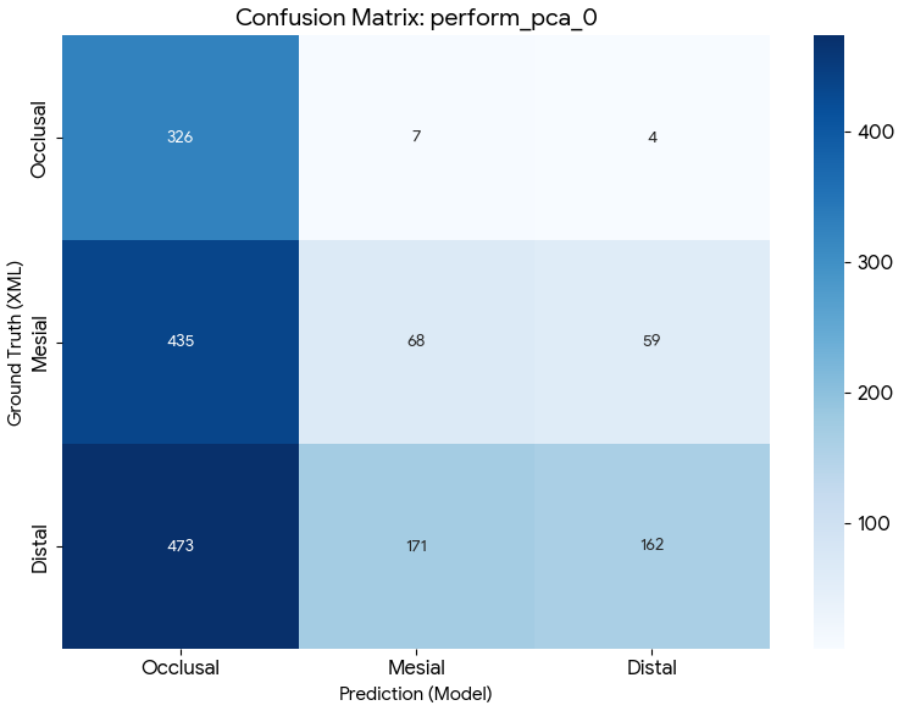
This baseline method uses Principal Component Analysis (PCA) to estimate the main axis of the tooth based on pixel distribution.

- **Macro F1-score:** 0.30
- **Key Issue:** Axis flipping in symmetric tooth shapes (e.g., molars)

```
=== perform_pca_0 ===
Confusion Matrix:
[[326   7   4]
 [435  68  59]
 [473 171 162]]
Report:
              precision    recall  f1-score   support

    Occlusal       0.26      0.94      0.41       347
      Mesial       0.28      0.12      0.17       577
      Distal       0.72      0.20      0.31       819

   micro avg       0.33      0.32      0.32      1743
   macro avg       0.42      0.42      0.30      1743
weighted avg       0.48      0.32      0.28      1743
```



Picture of Method 1: Standard PCA (Baseline) Evaluation

Method 2: Conditional Ratio

This method introduces a width-to-height ratio check to reduce incorrect rotations.

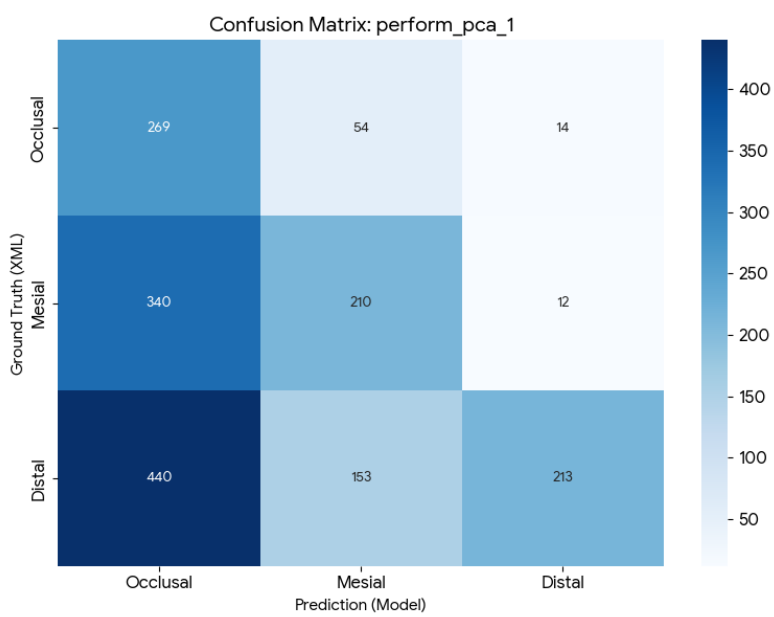
- **Macro F1-score:** 0.40
- **Improvement:** Reduces 90-degree misalignment in some cases

However, performance is still inconsistent across different tooth shapes.

```
=== perform_pca_1 ===
Confusion Matrix:
[[269  54  14]
 [340 210  12]
 [440 153 213]]
Report:
              precision    recall  f1-score   support

    Occlusal       0.26      0.78      0.39       347
      Mesial       0.50      0.36      0.42       577
      Distal       0.89      0.26      0.40       819

   micro avg       0.41      0.40      0.40      1743
   macro avg       0.55      0.47      0.40      1743
weighted avg       0.64      0.40      0.41      1743
```



Picture of Method 2: Conditional Ratio Evaluation

Method 3: Maximum Projected Span

This method selects the axis based on the maximum physical span of the tooth rather than statistical variance.

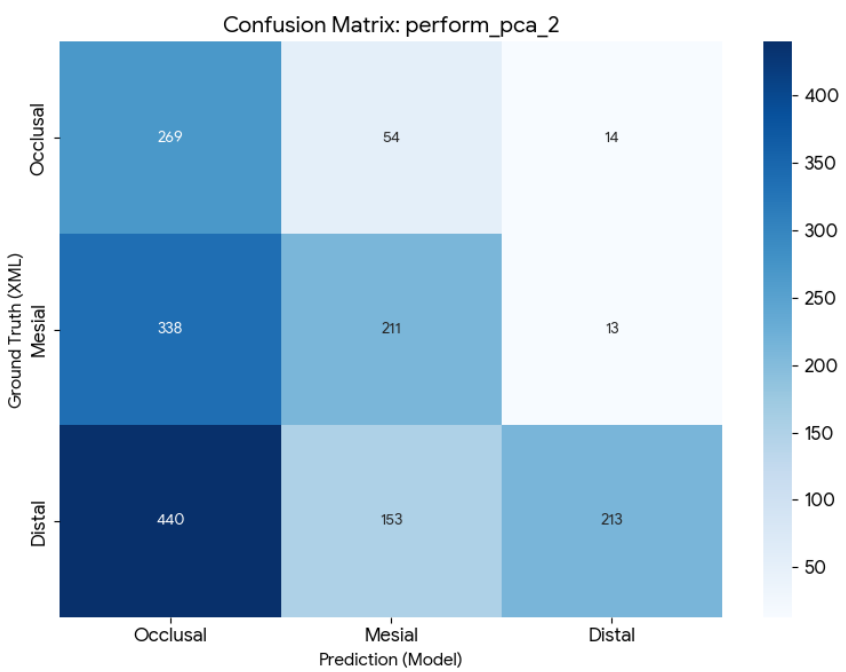
- **Macro F1-score:** 0.40
- **Improvement:** More stable than Conditional Ratio

It reduces sensitivity to pixel density but still lacks robustness in complex cases.

```
=== perform_pca_2 ===
Confusion Matrix:
[[269  54  14]
 [338 211  13]
 [440 153 213]]
Report:
              precision    recall  f1-score   support

    Occlusal       0.26      0.78      0.39       347
      Mesial       0.50      0.37      0.42       577
      Distal       0.89      0.26      0.40       819

   micro avg       0.41      0.40      0.40      1743
   macro avg       0.55      0.47      0.40      1743
weighted avg       0.64      0.40      0.41      1743
```



Picture of Method 3: Maximum Projected Span Evaluation

Method 4: Split-Centroid Anatomy

This method divides the tooth mask into upper and lower regions to estimate an anatomically consistent axis.

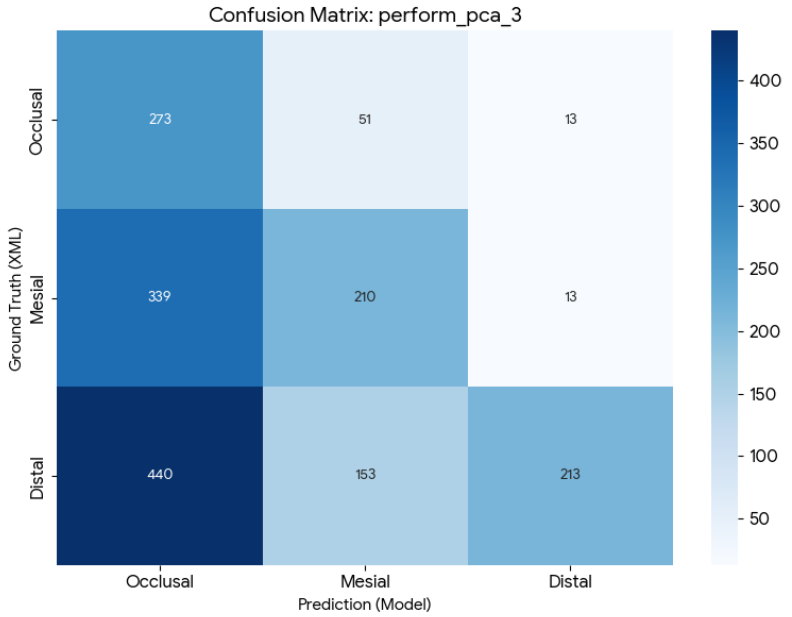
- **Macro F1-score:** 0.41 (Best in Phase 1)
- **Strength:** Robust to noise and symmetric shapes

This method provides the most stable orientation and is selected for further development.

```
=== perform_pca_3 ===
Confusion Matrix:
[[273  51  13]
 [339 210  13]
 [440 153 213]]
Report:
              precision    recall  f1-score   support

    Occlusal       0.26      0.79      0.39       347
      Mesial       0.51      0.36      0.42       577
      Distal       0.89      0.26      0.40       819

   micro avg       0.41      0.40      0.40      1743
   macro avg       0.55      0.47      0.41      1743
weighted avg       0.64      0.40      0.41      1743
```



Picture of Method 4: Split-Centroid Anatomy Evaluation

Method 5: Furthest Point Alignment (Concept)

This method determines orientation using the farthest point from the centroid.

- **Result:** Not selected
- **Issue:** Highly unstable when segmentation is incomplete

Method 6: Absolute Vertical Prior

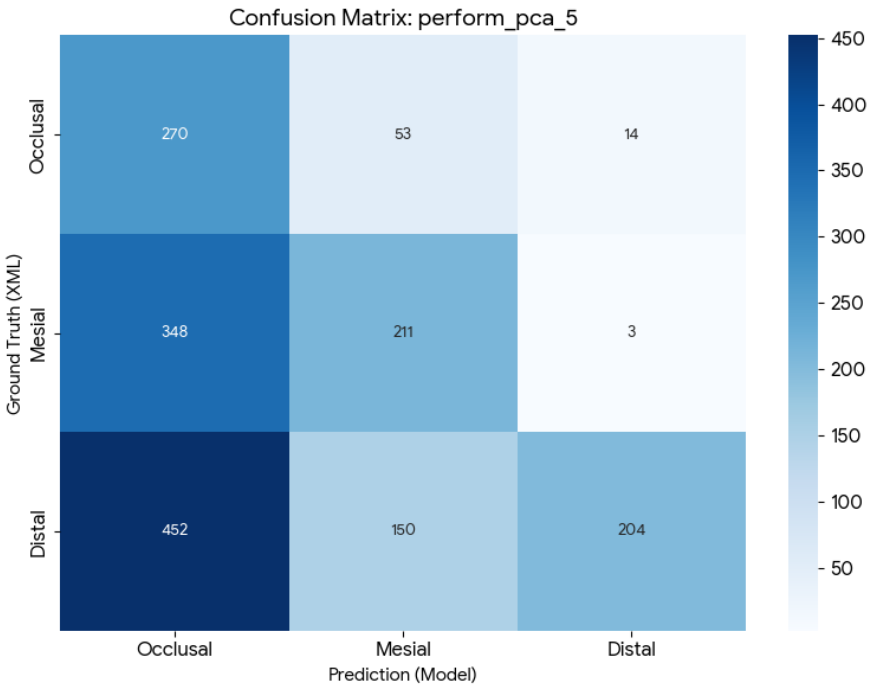
This method forces all teeth to align with the vertical axis of the image.

- **Macro F1-score:** 0.40
- **Limitation:** Sensitive to overall image tilt

```
=== perform_pca_5 ===
Confusion Matrix:
[[270  53  14]
 [348 211   3]
 [452 150 204]]
Report:
              precision    recall  f1-score   support

    Occlusal       0.25      0.78      0.38       347
      Mesial       0.51      0.37      0.43       577
      Distal       0.92      0.25      0.39       819

   micro avg       0.40      0.39      0.40      1743
   macro avg       0.56      0.46      0.40      1743
weighted avg       0.65      0.39      0.40      1743
```



Picture of Method 6: Absolute Vertical Prior Evaluation

Method 7: Multi-Class Segmentation (Concept)

This conceptual method proposes separating crown and root during model training.

- **Status:** Not implemented
- **Reason:** Requires re-annotation of the dataset

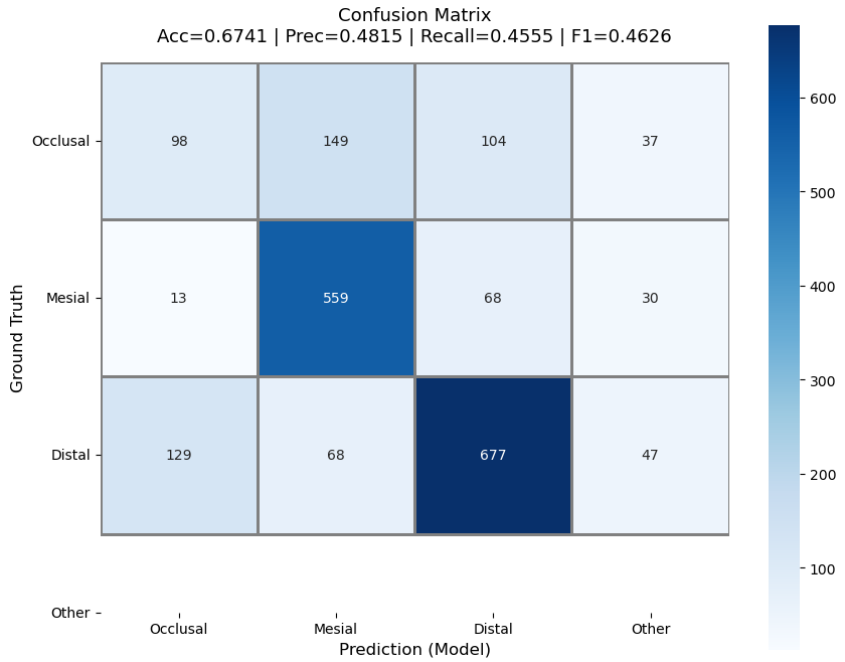
Phase 2: Optimized Classification Models

Method 1: X-Thirds Dominant Zone (Final Model)

This model divides the tooth into three vertical zones (40%–20%–40%) and applies multi-zone point-cloud voting.

- **Accuracy:** 70.54%
- **Macro F1-score:** 0.4714
- **Strength:** Balanced performance across classes

This version aligns well with the distribution of caries in the dataset, particularly for Mesial and Distal surfaces.

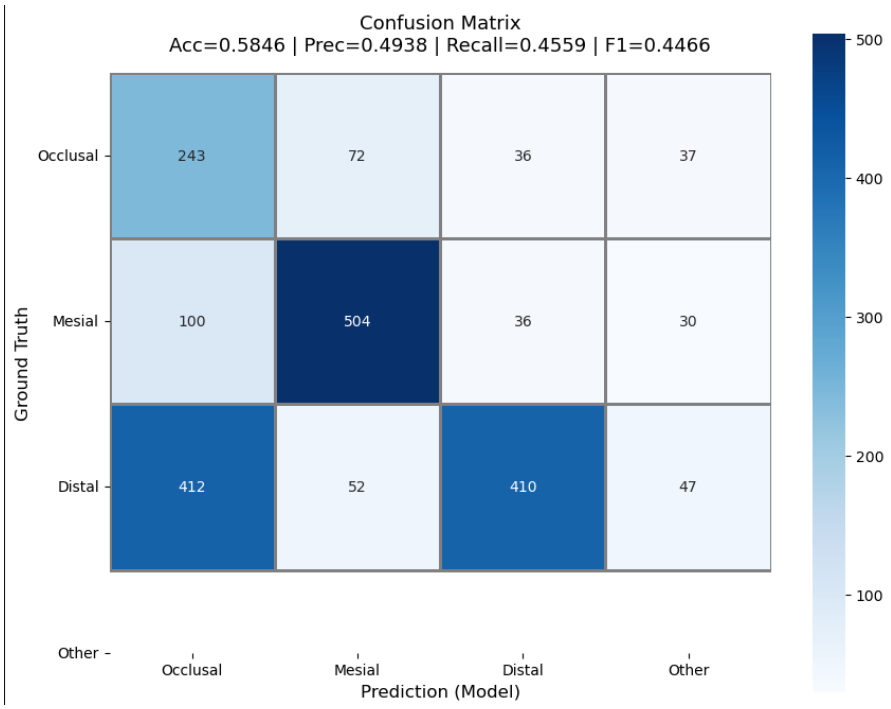


Picture of Method 1: X-Thirds Dominant Zone Evaluation

Method 2: Diagonal-from-Centroid

This model uses diagonal partitioning to improve Occlusal detection.

- **Accuracy:** 60.28%
- **Macro F1-score:** 0.4601
- **Strength:** Improved Occlusal recall
- **Weakness:** Significant drop in Distal performance



Picture of Method 2: X- Diagonal-from-Centroid Evaluation

Results & Insights

This section presents experimental results from both model evaluation and hypothesis testing. The objective is to determine whether the **X-Thirds Dominant Zone** or the **Diagonal-from-Centroid** approach should be selected as the final model.

Model Performance Summary

The evaluation was conducted on a dataset of **1,979 samples**, with the following results:

Model	Accuracy	Macro F1-score
X-Thirds Dominant Zone	70.54%	0.4714
Diagonal-from-Centroid	60.28%	0.4601

Although the **X-Thirds Dominant Zone** shows a significantly higher accuracy (~10%), the difference in Macro F1-score is relatively small (0.0113). This suggests that both models perform similarly when evaluated across all classes equally.

1.Hypothesis Testing

To support model selection, two statistical methods were applied:

- Bootstrap Resampling (for Macro F1-score)
- Two-Proportion Z-Test (for Accuracy)

Bootstrap Resampling (Macro F1-score)

Observed Results:

- F1 (X-Thirds Dominant Zone) = 0.4714
- F1 (Diagonal-from-Centroid) = 0.4601
- Observed Difference = 0.0113

Hypothesis Testing Result:

- 95% Confidence Interval: [-0.0091, 0.0307]
- P-value: 0.2820

Interpretation

The confidence interval includes zero, meaning that:

- In some examples, the Diagonal-from-Centroid model performs better
- In others, the X-Thirds Dominant Zone model performs better

Additionally, the P-value (0.2820) is greater than 0.05, indicating that:

There is **no statistically significant difference** between the two models in terms of Macro F1-score.

Conclusion (Bootstrap)

Both models have **comparable performance when evaluated across all classes**, particularly under data imbalance conditions.

2.Two-Proportion Z-Test (Accuracy)

$$Z = \frac{p_1 - p_2}{\sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Observed Results:

- X-Thirds Dominant Zone Accuracy = 70.54% (1396 / 1979)
- Diagonal-from-Centroid Accuracy = 60.28% (1193 / 1979)

Hypothesis Testing Result:

- **Z-statistic:** 6.7837
- **P-value:** 1.17×10^{-11}

3.Interpretation

The P-value is extremely small (<< 0.05), which indicates:

The difference in accuracy is **statistically significant** and not due to random chance.

Conclusion (Z-Test)

The **X-Thirds Dominant Zone** model performs significantly better than the **Diagonal-from-Centroid** model in terms of overall accuracy.

Combined Interpretation

The two statistical tests provide different perspectives:

- **Bootstrap (F1-score):**

  ⟶ Both models perform similarly when treating all classes equally

- **Z-Test (Accuracy):**

  ⟶ X-Thirds Dominant Zone performs significantly better overall

This difference occurs because:

- The dataset is **imbalanced** (Mesial and Distal dominate)
- Accuracy is influenced more by dominant classes
- Macro F1-score treats all classes equally

4.Error Analysis and Trade-off

From the confusion matrix:

- **X-Thirds Dominant Zone**
    - Strong performance in Mesial and Distal
    - Lower recall in Occlusal
- **Diagonal-from-Centroid**
    - Significantly improves Occlusal detection
    - Causes a major drop in Distal performance

Root Cause

The issue arises from geometric design:

- The Diagonal-from-Centroid method divides the tooth using diagonal lines
- However, due to **elongated bounding boxes (including root regions)**:
    - Occlusal zone becomes disproportionately large
    - Distal zone becomes too narrow

This leads to frequent misclassification of Distal lesions as Occlusal.

5.Final Decision

Based on both statistical testing and error analysis:

- The X-Thirds Dominant Zone model achieves significantly higher accuracy
- Macro F1-score differences are not statistically significant
- The X-Thirds Dominant Zone model provides better stability and fewer trade-offs

Therefore, the **X-Thirds Dominant Zone** approach is selected as the final model for this project.

Limitations & Challenges

Despite the improvements achieved by the X-Thirds Dominant Zone model, several limitations exist in the dataset that directly impact model performance and evaluation.

1.Class Imbalance

The dataset is highly imbalanced, with Mesial and Distal cases dominating most samples, while Occlusal cases are relatively fewer.

- This leads to:
    - Higher overall accuracy driven by dominant classes
    - Lower sensitivity to minority classes (especially Occlusal)

This explains why the Diagonal-from-Centroid method improves Occlusal detection but still underperforms overall.

2.Inclusion of Root Region in Bounding Box

The bounding boxes used for each tooth include both crown and root regions.

- This creates:
    - Vertically elongated shapes
    - Distortion in geometric zoning (especially diagonal-based methods)

As observed in Section Results & Insights, this directly affects the Diagonal-from-Centroid model by enlarging the Occlusal zone and shrinking the Distal zone.

3.Dependence on Segmentation Quality

The dataset relies on outputs from instance segmentation models.

- Issues such as:
    - Incomplete masks
    - Missing root or crown regions
    - Noise from adjacent teeth

can propagate errors into later stages (PCA and classification).

4.Lack of Anatomical Separation (Crown vs Root)

The dataset does not explicitly separate crown and root regions.

- These limits:
    - Accurate representation of clinically relevant areas
    - The effectiveness of geometry-based methods

This is a key limitation preventing more advanced zoning strategies from performing optimally.

5.Limited Data Diversity

The dataset is collected from a specific source (faculty-provided dataset), which may result in:

- Limited variation in:
    - Image orientation
    - Patient characteristics
    - Tooth morphology

This may reduce the generalizability of the model to real-world scenarios.

6.Evaluation Constraints

The evaluation is performed on a single dataset split without external validation.

- This introduces:
    - Risk of overfitting to dataset characteristics
    - Limited confidence in generalization performance

Recommendations & Next Steps

Based on the identified dataset limitations, the following improvements are proposed.

1.Data Balancing

To address class imbalance:

- Collect more Occlusal samples
- Apply data augmentation techniques

This will improve model fairness across all classes and reduce bias toward dominant classes.

2.Crown-Root Separation

A key improvement is to separate crown and root regions before classification.

- This will:
    - Normalize tooth shape
    - Improve geometric consistency
    - Enable more accurate zoning

This is especially important for improving the Diagonal-from-Centroid approach.

3.Improved Annotation Quality

Enhancing annotation quality will directly improve model performance:

- Refine segmentation masks
- Ensure consistent labeling across samples
- Reduce noise from adjacent structures

4.Adaptive Zoning Methods

Instead of fixed geometric rules:

- Develop adaptive zoning based on tooth shape
- Or use learning-based approaches

This would reduce sensitivity to bounding box distortion.

5.Integration of Learning-Based Models

Future work can be incorporated:

- Deep learning classifiers
- Hybrid models combining geometry and learned features

This will allow the system to capture more complex spatial patterns.

6.External Validation

To improve reliability:

- Test the model on external datasets
- Perform cross-validation

This will ensure that the model generalizes beyond the current dataset.

Conclusion

This project presents a complete pipeline for automated tooth surface classification from panoramic dental X-ray images, combining detection, segmentation, and geometric analysis techniques.

In Phase 1, multiple PCA-based methods were explored to improve tooth orientation estimation, where the Split-Centroid approach demonstrated the highest robustness. In Phase 2, the focus shifted toward optimizing surface classification through zoning strategies, leading to the development of two main approaches: **X-Thirds Dominant Zone** and **Diagonal-from-Centroid**.

Experimental results on 1,979 samples show that the X-Thirds Dominant Zone achieves higher overall accuracy (70.54%) compared to the Diagonal-from-Centroid method (60.28%). Statistical validation further confirms that this difference is significant in terms of accuracy (Two-Proportion Z-Test, $p < 0.05$), while no significant difference is observed in Macro F1-score (Bootstrap, $p > 0.05$), indicating similar performance when class balance is considered.

A detailed error analysis reveals that the Diagonal-from-Centroid approach improves detection of Occlusal surfaces but introduces significant errors in Distal classification due to geometric distortion caused by elongated bounding boxes. This highlights an important insight that model performance depends not only on algorithm design but also on how well the method aligns with data characteristics and anatomical structure.

Overall, the X-Thirds Dominant Zone is selected as the final model due to its higher accuracy, better stability, and lower trade-off across classes. The findings of this study demonstrate that combining geometric reasoning with statistical validation provides a reliable framework for developing interpretable and effective medical imaging solutions.

References & Appendix

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GitHub

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